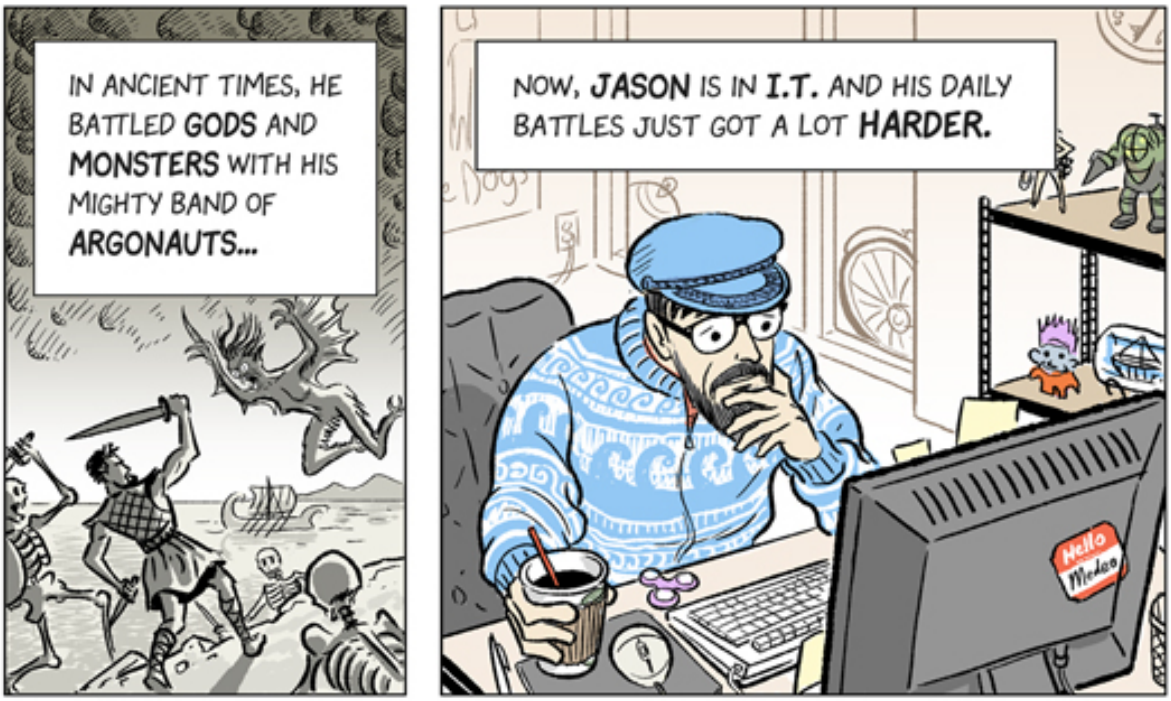
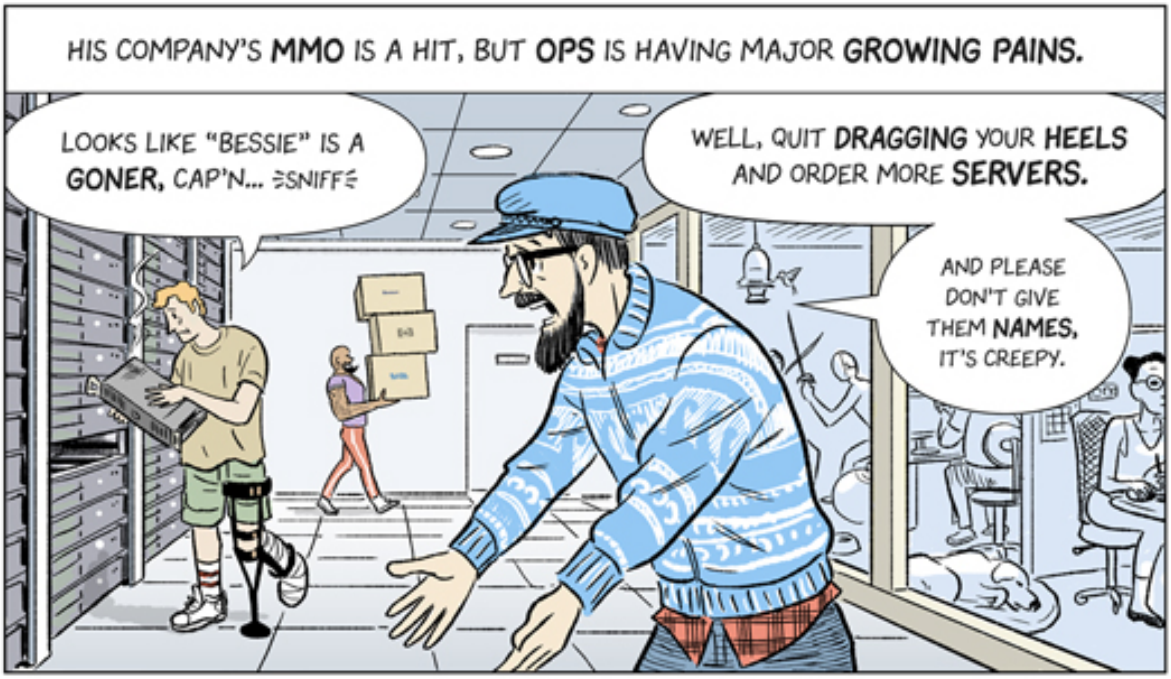
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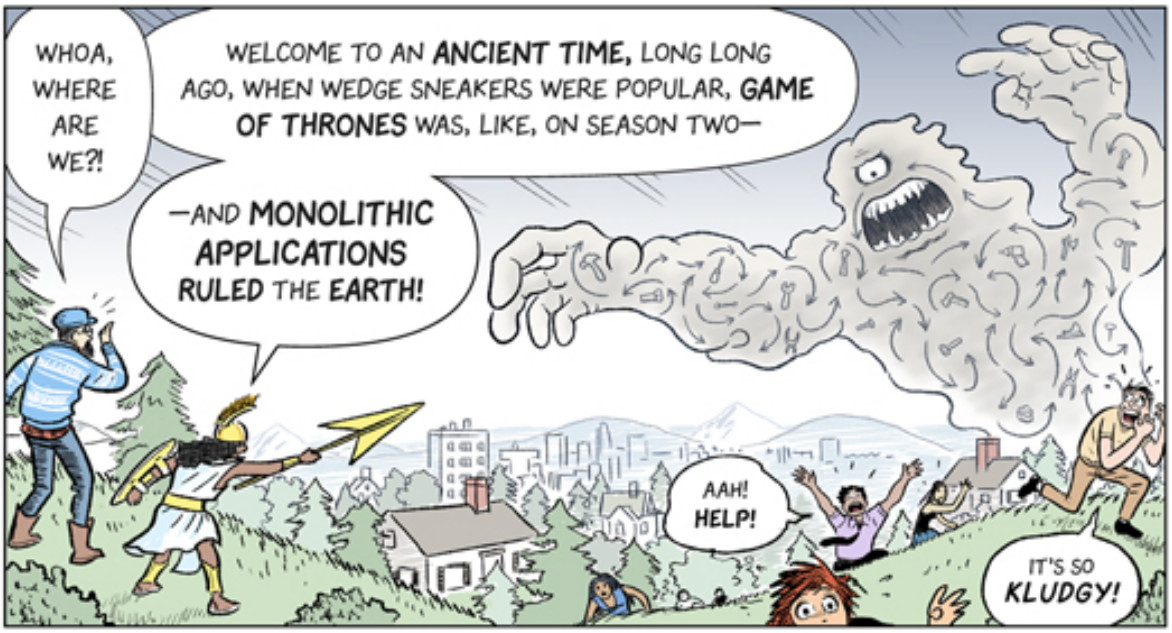



HIS COMPANY'S MMO IS A HIT, BUT OPS IS HAVING MAJOR GROWING PAINS.
LOOKS LIKE "BESSIE" IS A GONER, CAP'N... :SNIFF:
WELL, QUIT DRAGGING YOUR HEELS AND ORDER MORE SERVERS.
AND PLEASE DON'T GIVE THEM NAMES, IT'S CREEPY.


FOR HERA'S SAKE, WHO KNEW DELIVERING THESE SERVICES WOULD BE SO HARD?
WHY ISN'T MY CODE UP YET??
OUR UPDATE CYCLES ARE TOO SLOW!
WE CAN'T SCALE UP FAST ENOUGH!
GAME'S GONNA GET STALE, BOSS...
SORRY TO CALL AT 3:00AM, BUT—
OH, CRAP— POWER'S OUT AGAIN!
WHY CAN'T IT JUST WORK ALL THE TIME?!


IT CAN—
—WITH KUBERNETES!
HUH?
WHO—?
IT'S ME. ATHENA! GODDESS OF WISDOM, CRAFT, AND CONTAINERIZED APPLICATIONS.
I THOUGHT IT WAS "WISDOM, CRAFT, AND WAR."
MORTALS HAVE ENOUGH WAR. YOU NEED CONTAINERS!
ALSO, I'M DIVERSIFYING.

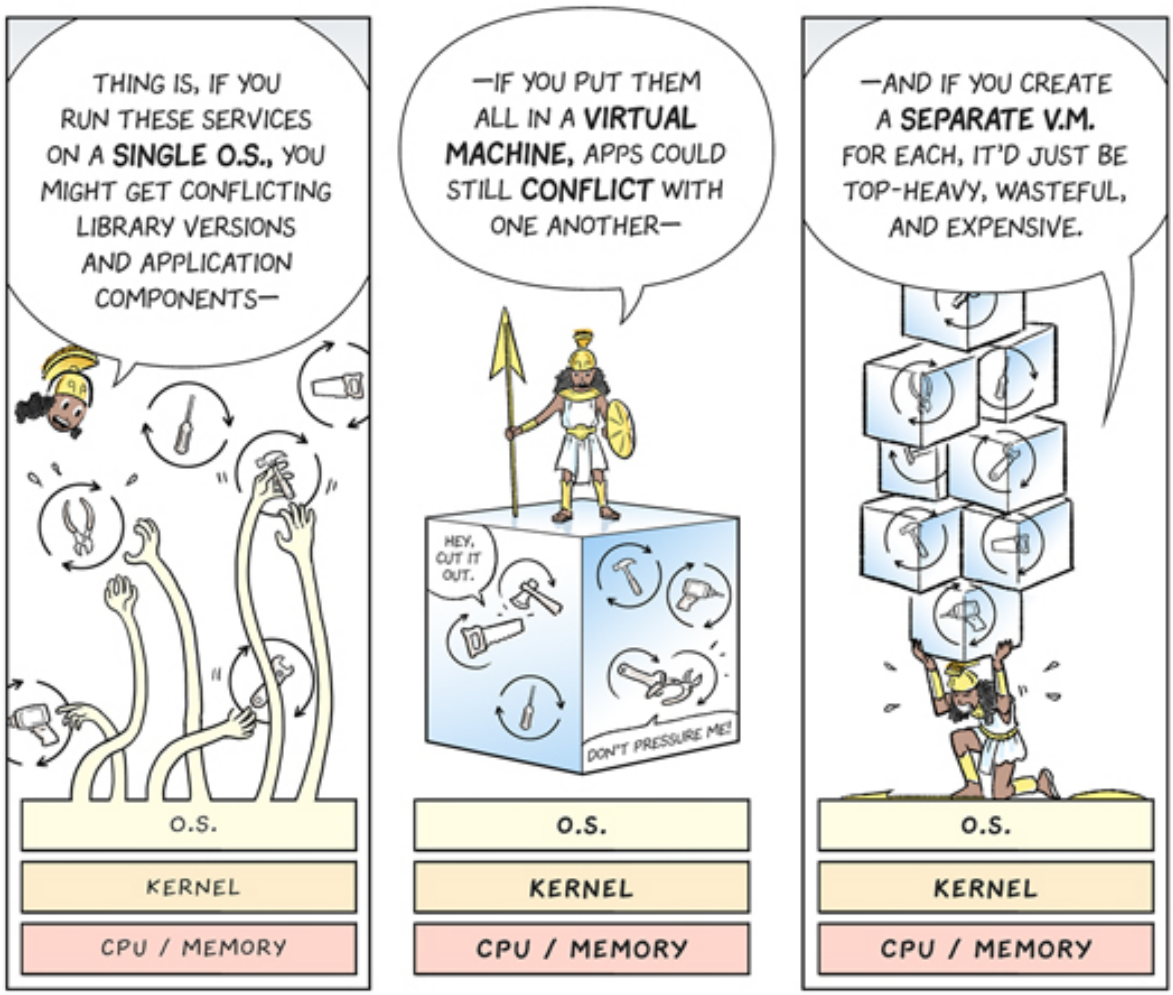

WELL, WE'RE ALREADY DEPLOYING APPS IN CONTAINERS, BUT IT'S HARDLY FIXED EVERYTHING...
MANAGING THEM IS STILL SLOW, INEFFICIENT, AND FULL OF HOLES.
PUTTING APPS IN CONTAINERS IS A GREAT FIRST STEP—
—BUT NOW YOU NEED TO ORCHESTRATE THOSE PUPPIES!
POKE! POKE!
THAT'S WHERE KUBERNETES CAN HELP.
BUT FIRST A LITTLE HISTORY LESSON.
FOLLOW ME!
HEY, IF YOU'RE A GREEK GODDESS, WHY DO YOU LOOK LIKE A JUNIOR COMIC-CON COSPLAYER?
SHUT UP, THAT'S WHY!


WHOA, WHERE ARE WE?!
WELCOME TO AN ANCIENT TIME, LONG LONG AGO, WHEN WEDGE SNEAKERS WERE POPULAR, GAME OF THRONES WAS, LIKE, ON SEASON TWO—
—AND MONOLITHIC APPLICATIONS RULED THE EARTH!
AAH! HELP!
IT'S SO KLUDGY!


MONOLITHS HAD IMPRESSIVE FEATURE SETS, BUT TOO MANY INTERDEPENDENT PARTS, SO INTEGRATION AND DEPLOYMENT WERE A NIGHTMARE.


BY CHOPPING THEM INTO INDIVIDUAL PROCESSES, THOUGH, WE'VE LEARNED TO BRING THESE MONSTERS UNDER CONTROL!
AARGH!
WHACK!

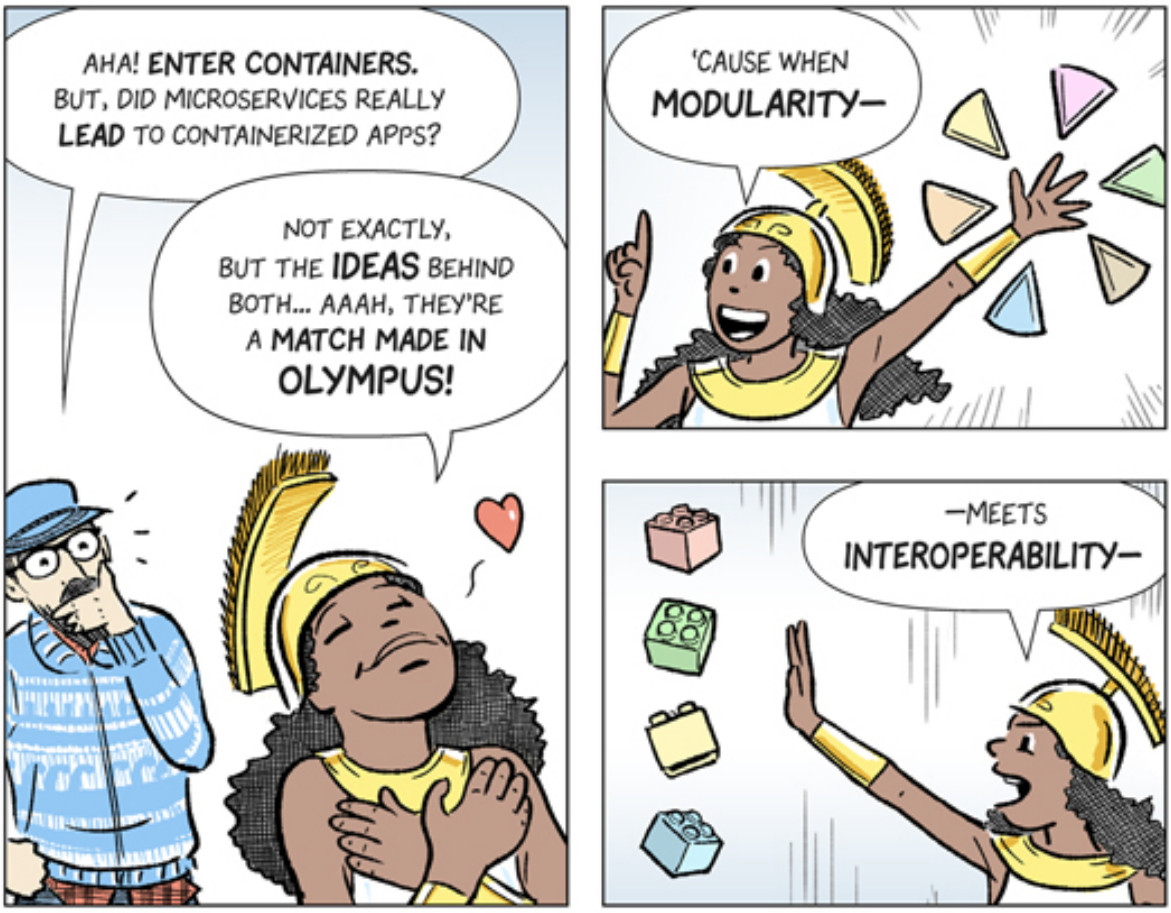

EACH ONE OF THESE "MICROSERVICES" CAN NOW BE DEBUGGED, UPDATED, AND DEPLOYED INDIVIDUALLY WITHOUT THE WHOLE PROJECT COMING TO A STANDSTILL.
AN IMPORTANT STEP ON THE PATH TO CONTINUOUS INTEGRATION AND DELIVERY.
THEY STILL HAVE TO TALK TO EACH OTHER RIGHT?
SURE! THEY'RE LOOSELY COUPLED, WITH A LIGHTWEIGHT PROTOCOL. JUST ENOUGH TO COOPERATE—
LOOKIN' SHARP.
YOU HITTIN' ON ME?
—AND TO PRESERVE THE NETWORK OF DEPENDENCIES THAT TIED THE BIG APPS TOGETHER.


THING IS, IF YOU RUN THESE SERVICES ON A SINGLE O.S., YOU MIGHT GET CONFLICTING LIBRARY VERSIONS AND APPLICATION COMPONENTS—
O.S.
KERNEL
CPU / MEMORY
—IF YOU PUT THEM ALL IN A VIRTUAL MACHINE, APPS COULD STILL CONFLICT WITH ONE ANOTHER—
HEY, CUT IT OUT.
DON'T PRESSURE ME!
O.S.
KERNEL
CPU / MEMORY
—AND IF YOU CREATE A SEPARATE V.M. FOR EACH, IT'D JUST BE TOP-HEAVY, WASTEFUL, AND EXPENSIVE.
O.S.
KERNEL
CPU / MEMORY

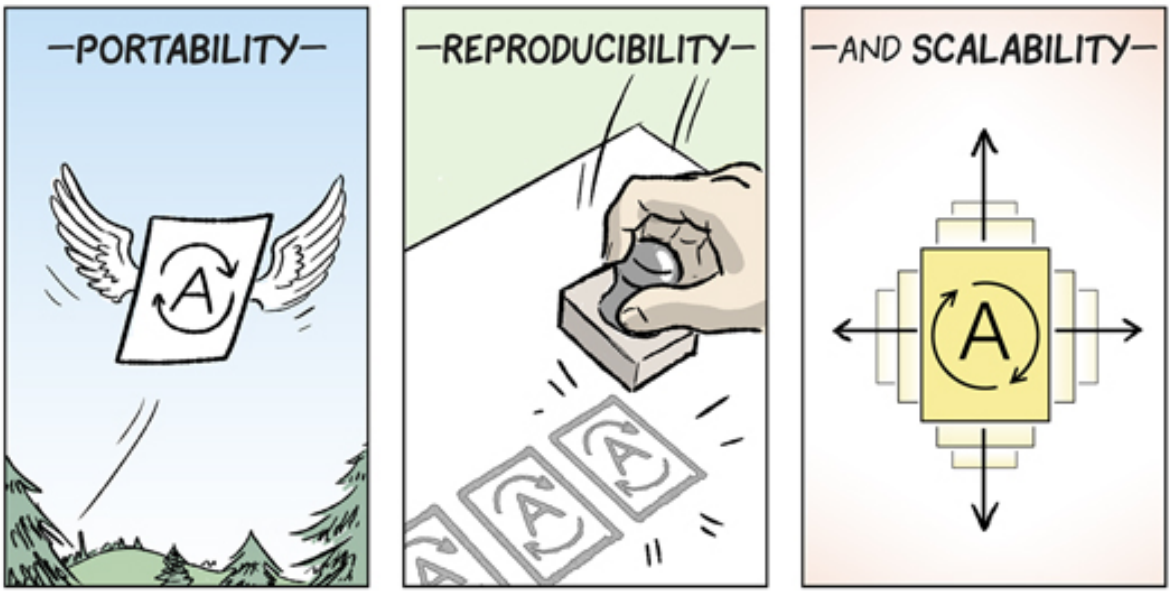

BUT, WHAT IF EACH PROCESS COULD SOMEHOW BE BUILT TO RUN ON ITS OWN—
—WITH ITS LIBRARIES AND SETTINGS BAKED IN?
A
IN SHORT, EVERYTHING EACH PROCESS NEEDS TO RUN ON ANY MACHINE, VIRTUAL OR BARE METAL.

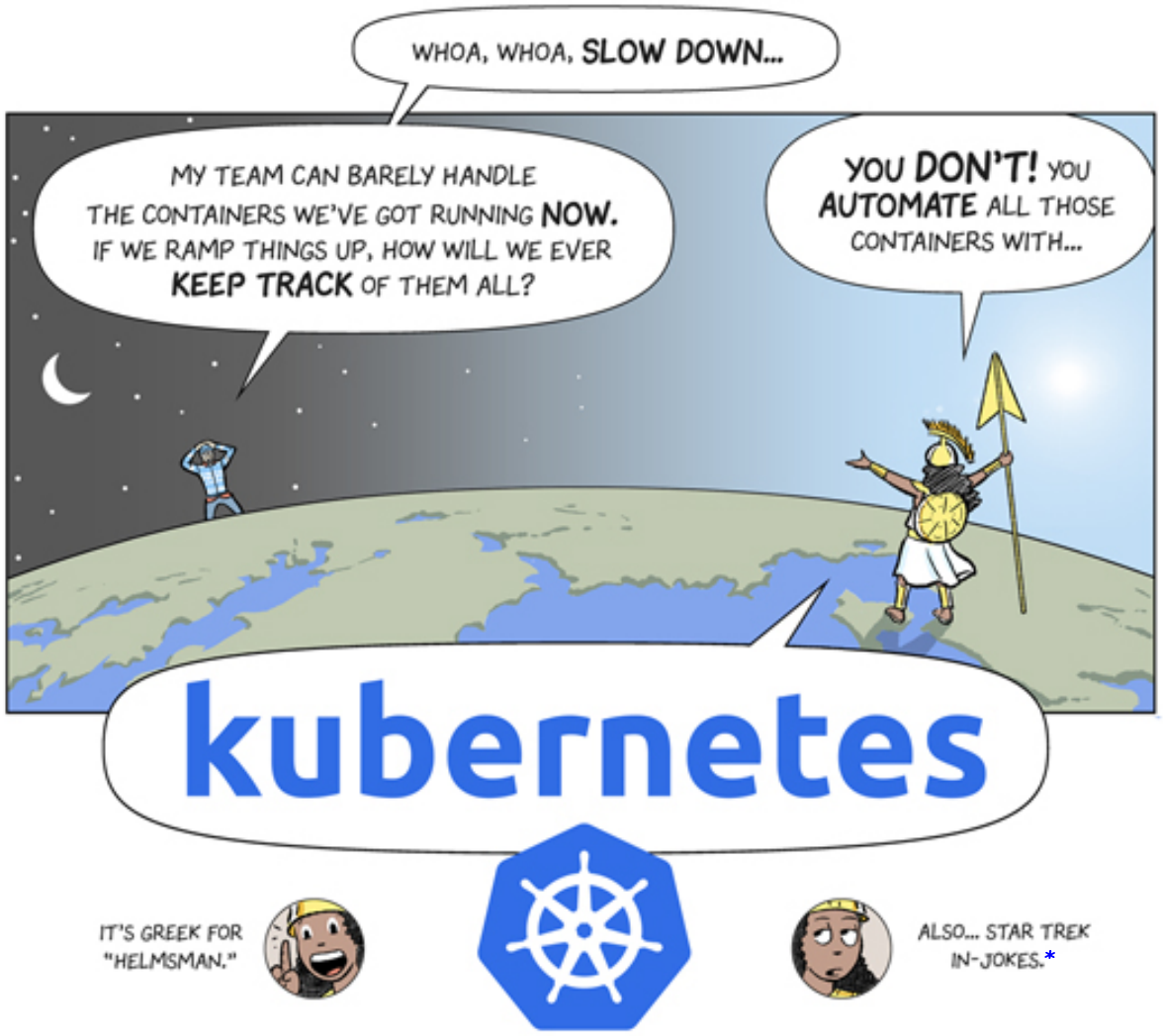

A
A
A

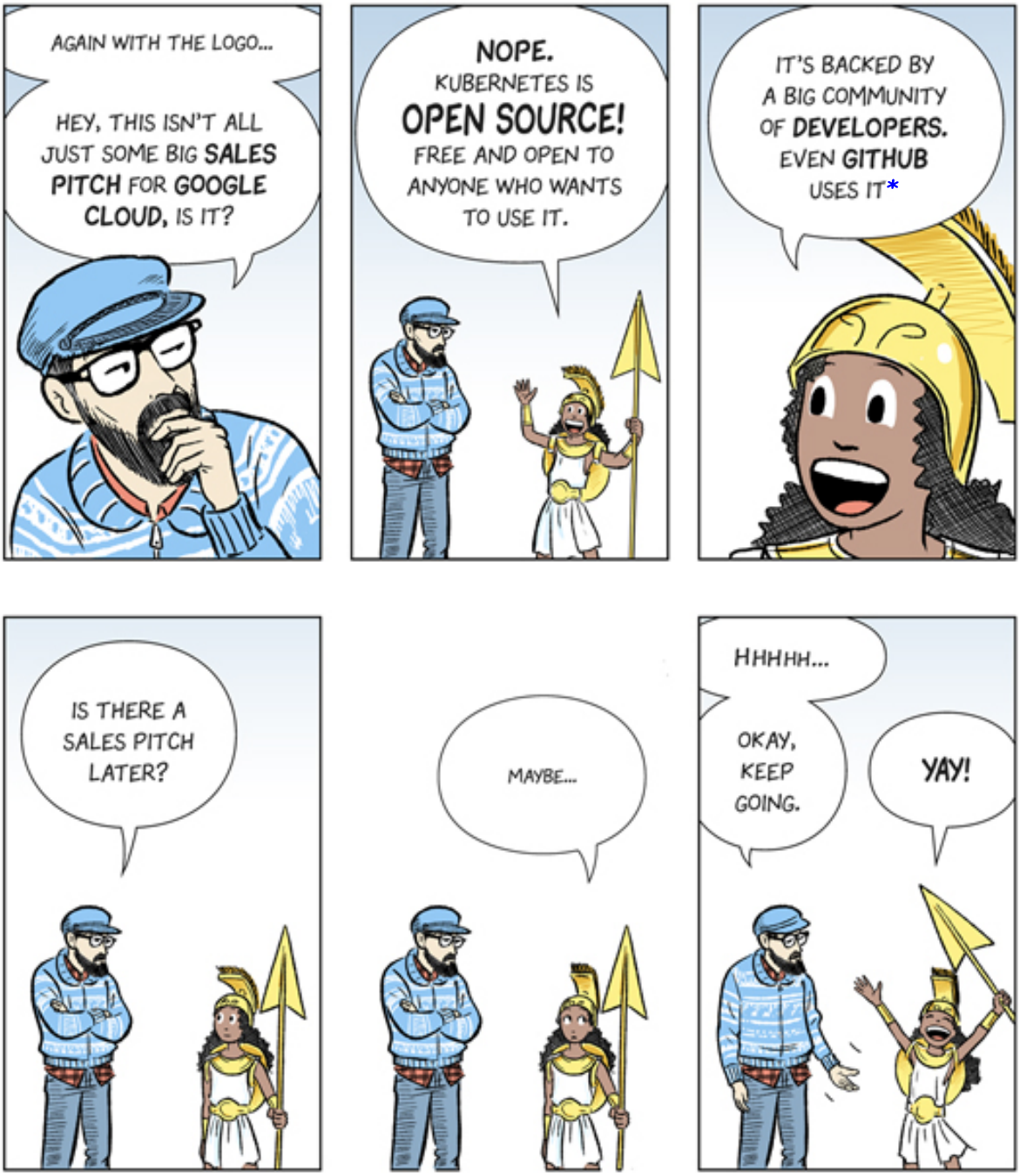
A **SELF-CONTAINED PROCESS.**

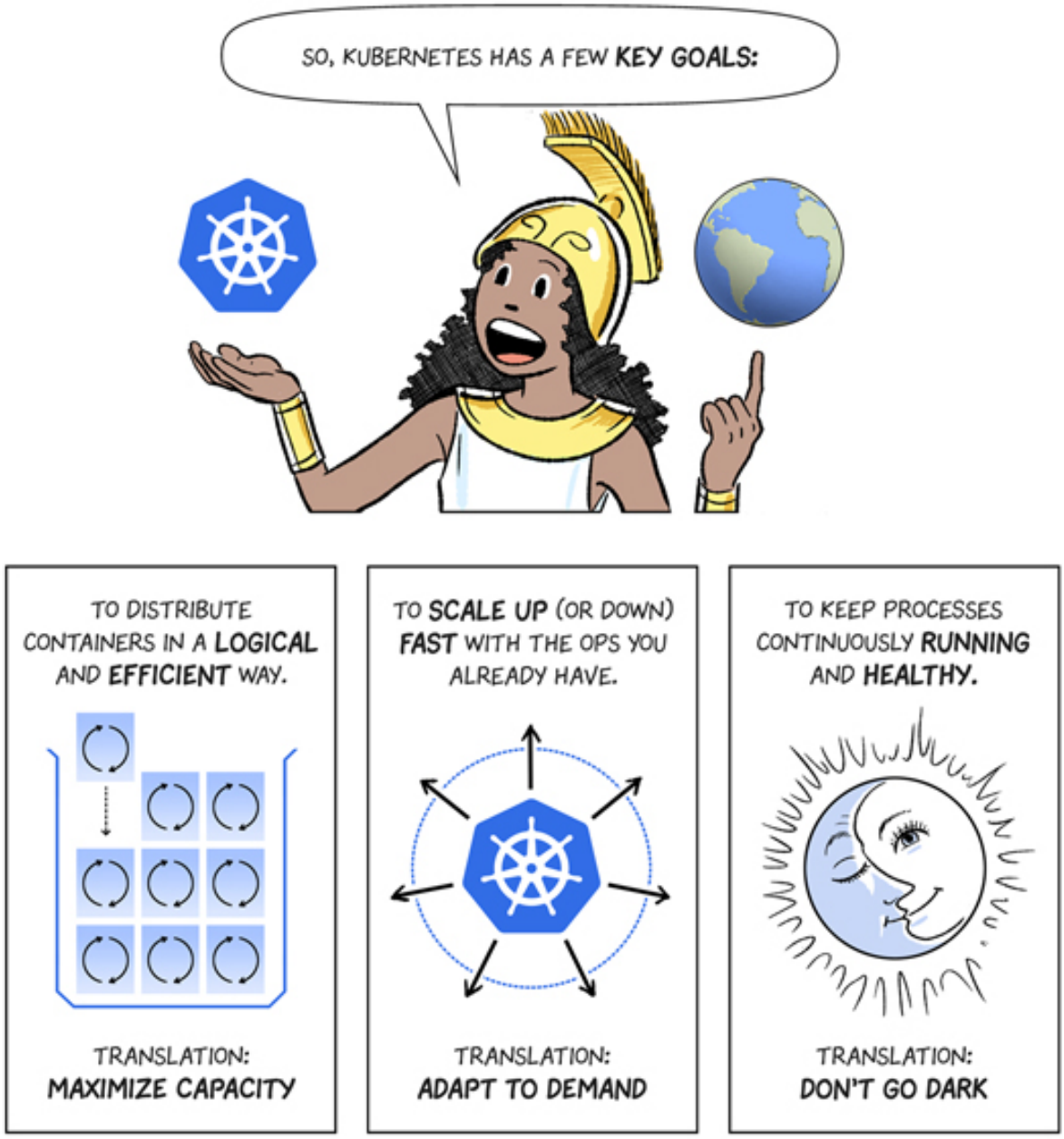


—YOU CAN START EXPLOITING **WHOLE NEW LEVELS** OF—

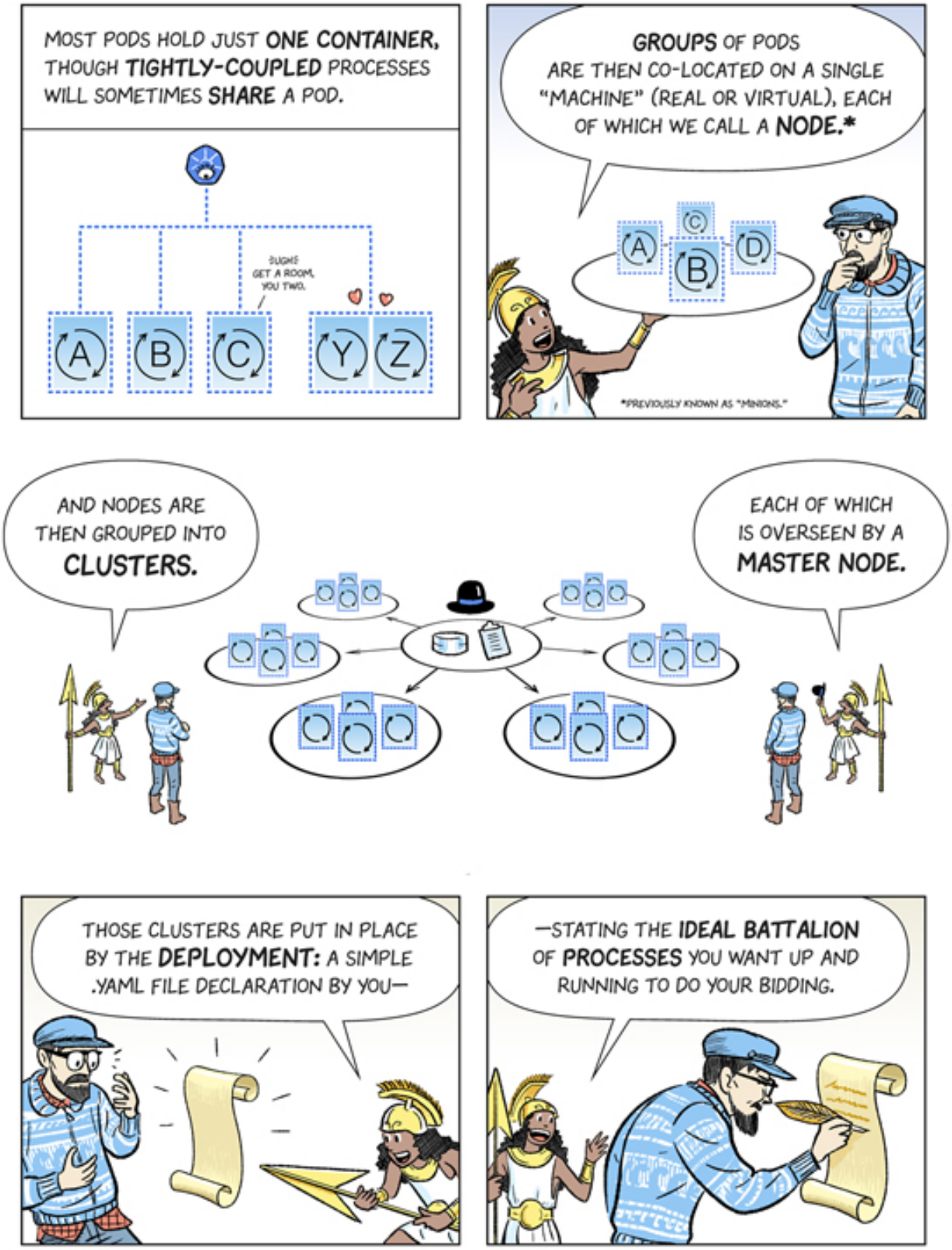

—PORTABILITY—
—REPRODUCIBILITY—
—AND SCALABILITY—
A
—ANYWHERE AND ANYTIME ACROSS THE WORLD!

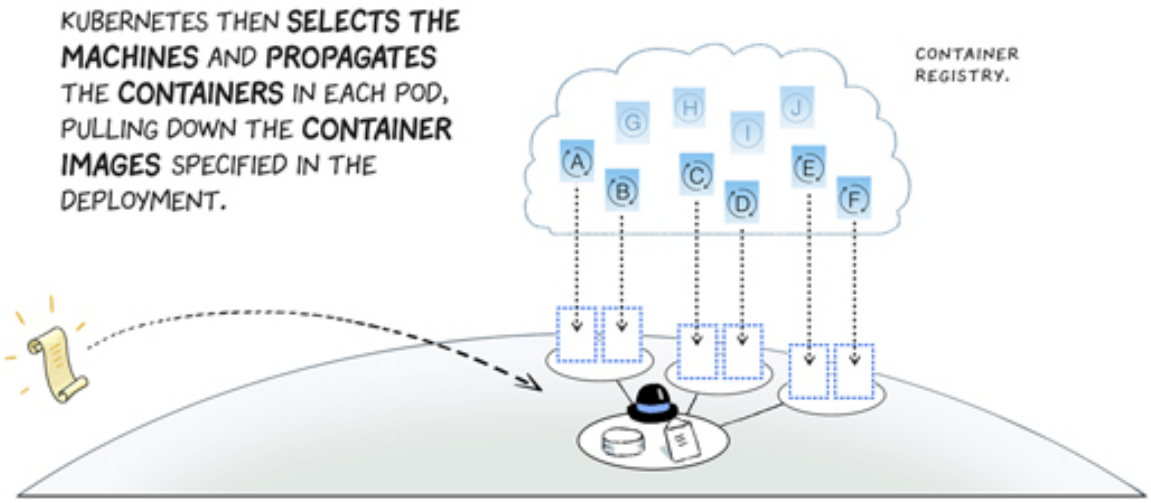

WHOA, WHOA, SLOW DOWN...
MY TEAM CAN BARELY HANDLE THE CONTAINERS WE'VE GOT RUNNING NOW. IF WE RAMP THINGS UP, HOW WILL WE EVER KEEP TRACK OF THEM ALL?
YOU DON'T! YOU AUTOMATE ALL THOSE CONTAINERS WITH...
kubernetes
IT'S GREEK FOR "HELMSMAN."
ALSO... STAR TREK IN-JOKES.*

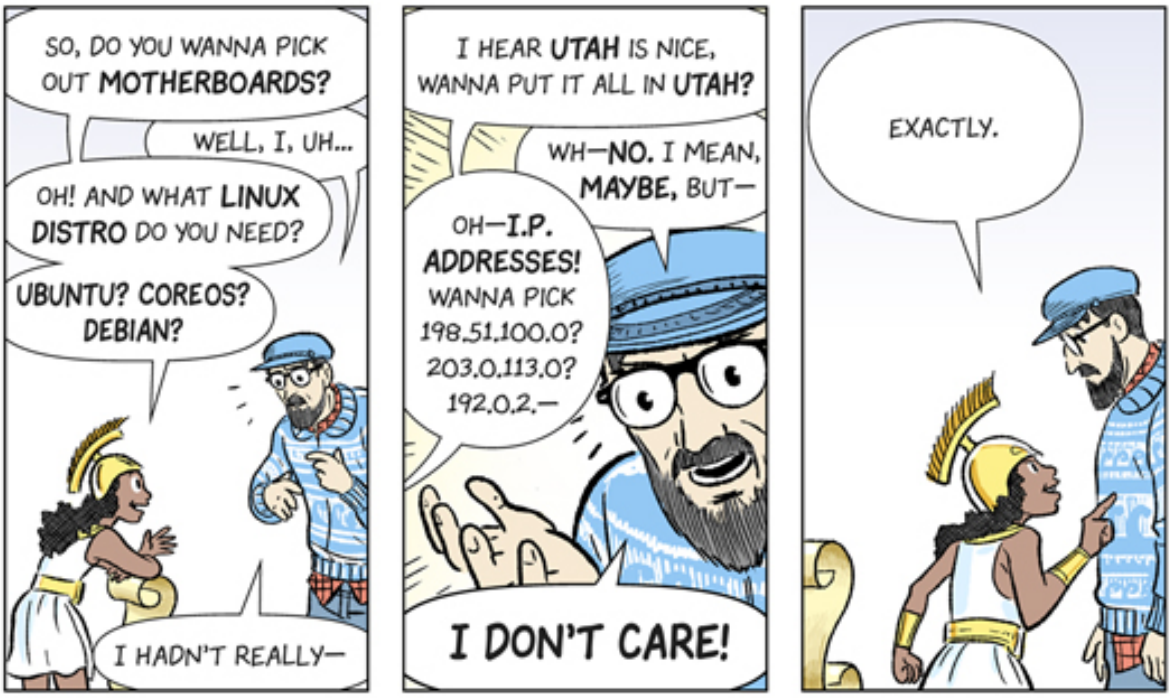

AGAIN WITH THE LOGO...
HEY, THIS ISN'T ALL JUST SOME BIG SALES PITCH FOR GOOGLE CLOUD, IS IT?
NOPE. KUBERNETES IS OPEN SOURCE! FREE AND OPEN TO ANYONE WHO WANTS TO USE IT.
IT'S BACKED BY A BIG COMMUNITY OF DEVELOPERS. EVEN GITHUB USES IT*
IS THERE A SALES PITCH LATER?
MAYBE...
HHHHH...
OKAY, KEEP GOING.
YAY!

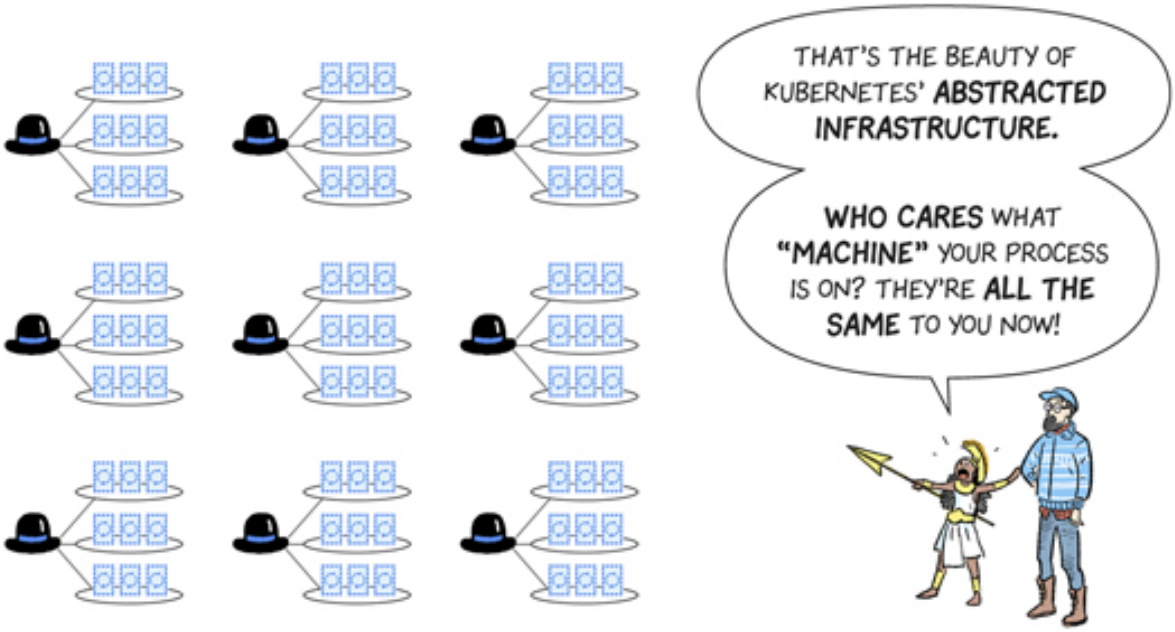

SO, KUBERNETES HAS A FEW KEY GOALS:
TO DISTRIBUTE CONTAINERS IN A LOGICAL AND EFFICIENT WAY.
TRANSLATION: MAXIMIZE CAPACITY
TO SCALE UP (OR DOWN) FAST WITH THE OPS YOU ALREADY HAVE.
TRANSLATION: ADAPT TO DEMAND
TO KEEP PROCESSES CONTINUOUSLY RUNNING AND HEALTHY.
TRANSLATION: DON'T GO DARK

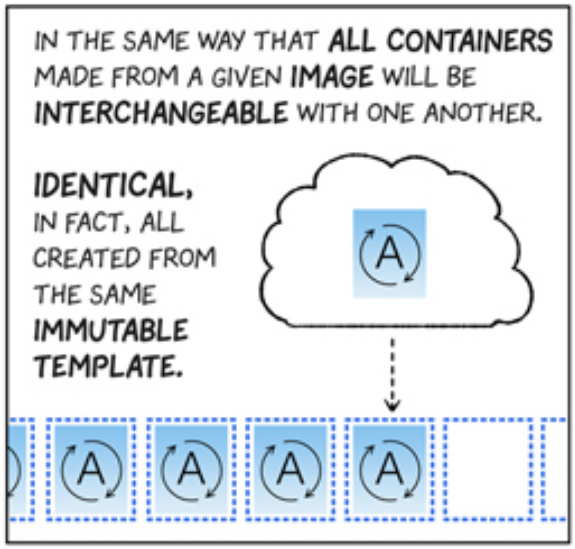

AND MOST IMPORTANTLY—
—TO GIVE YOU POWER OVER WHAT GETS DONE—
—WITHOUT FORCING YOU TO MICRO-MANAGE HOW.
TRANSLATION: OMG WEEKENDS
TO DO THIS, KUBERNETES ADDS A NEW LEVEL OF ABSTRACTION FOR CONTAINER MANAGEMENT:
THE POD.
PODS ARE THE SMALLEST BUILDING BLOCK IN THE KUBERNETES OBJECT MODEL. THE POD "SEES" THE CONTAINER—
—BUT KUBERNETES ONLY SEES THE POD.


MOST PODS HOLD JUST ONE CONTAINER, THOUGH TIGHTLY-COUPLED PROCESSES WILL SOMETIMES SHARE A POD.
A
B
C
SIGH! GET A ROOM, YOU TWO.
Y
Z
GROUPS OF PODS ARE THEN CO-LOCATED ON A SINGLE "MACHINE" (REAL OR VIRTUAL), EACH OF WHICH WE CALL A NODE.*
A
B
C
D
*PREVIOUSLY KNOWN AS "MINIONS."
AND NODES ARE THEN GROUPED INTO CLUSTERS.
EACH OF WHICH IS OVERSEEN BY A MASTER NODE.
THOSE CLUSTERS ARE PUT IN PLACE BY THE DEPLOYMENT: A SIMPLE .YAML FILE DECLARATION BY YOU—
—STATING THE IDEAL BATTALION OF PROCESSES YOU WANT UP AND RUNNING TO DO YOUR BIDDING.

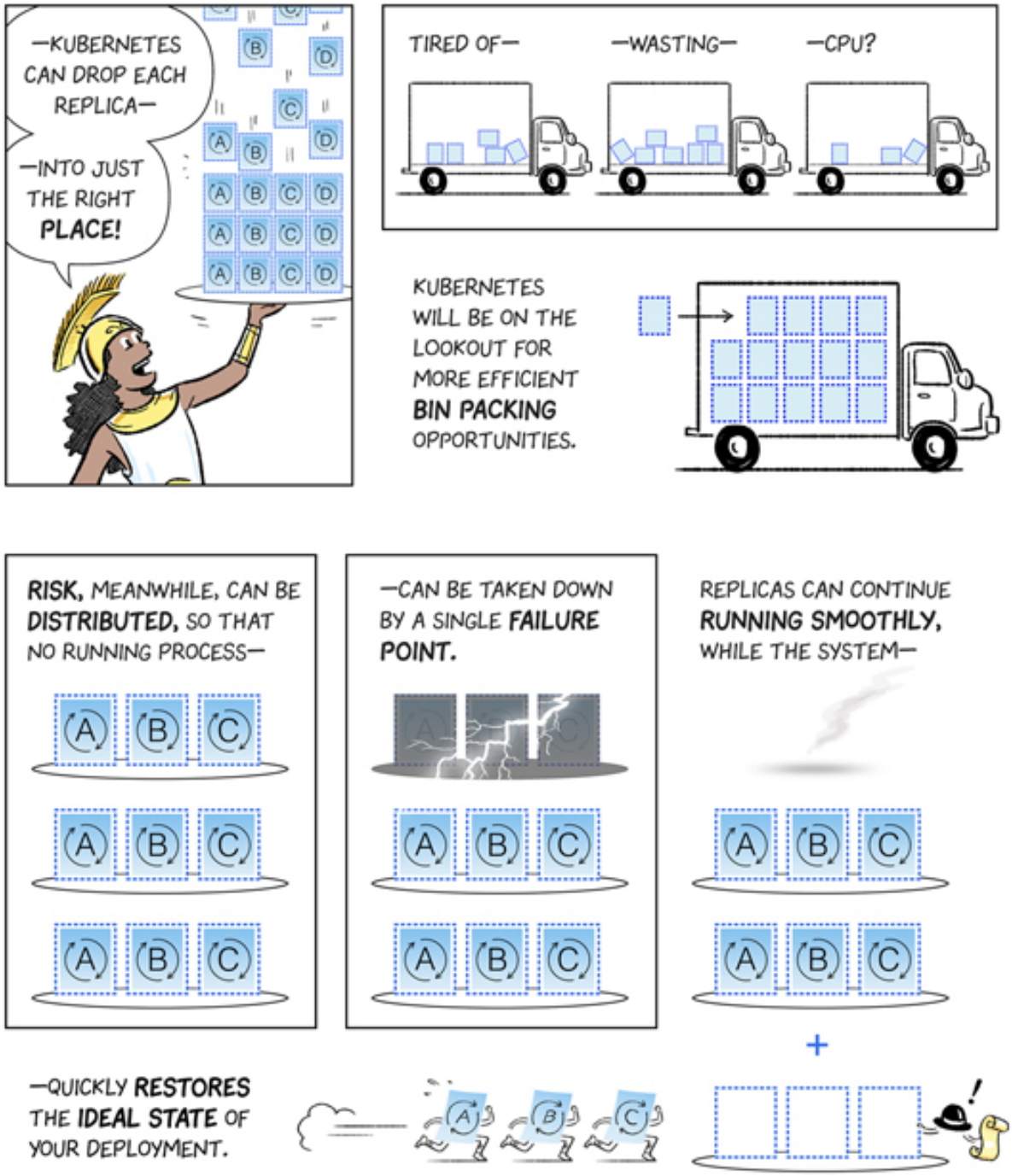

KUBERNETES THEN SELECTS THE MACHINES AND PROPAGATES THE CONTAINERS IN EACH POD, PULLING DOWN THE CONTAINER IMAGES SPECIFIED IN THE DEPLOYMENT.
CONTAINER REGISTRY.
A
B
C
D
E
F
G
H
I
J

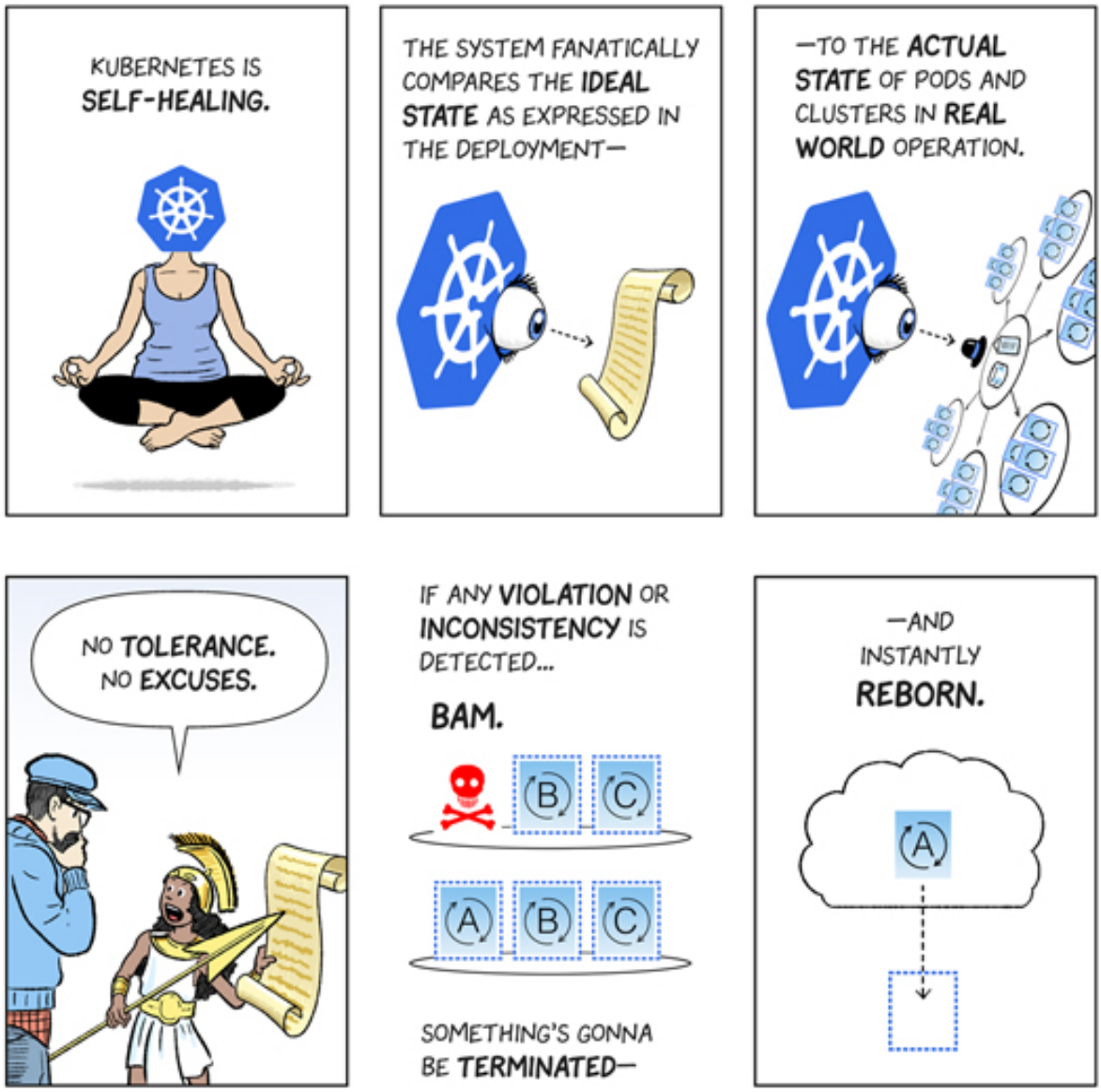

SO, DO YOU WANNA PICK OUT MOTHERBOARDS?
WELL, I, UH...
OH! AND WHAT LINUX DISTRO DO YOU NEED?
UBUNTU? COREOS? DEBIAN?
I HADN'T REALLY—
I HEAR UTAH IS NICE, WANNA PUT IT ALL IN UTAH?
WH—NO. I MEAN, MAYBE, BUT—
OH—I.P. ADDRESSES! WANNA PICK 198.51.100.0? 203.0.113.0? 192.0.2.—
I DON'T CARE!
EXACTLY.

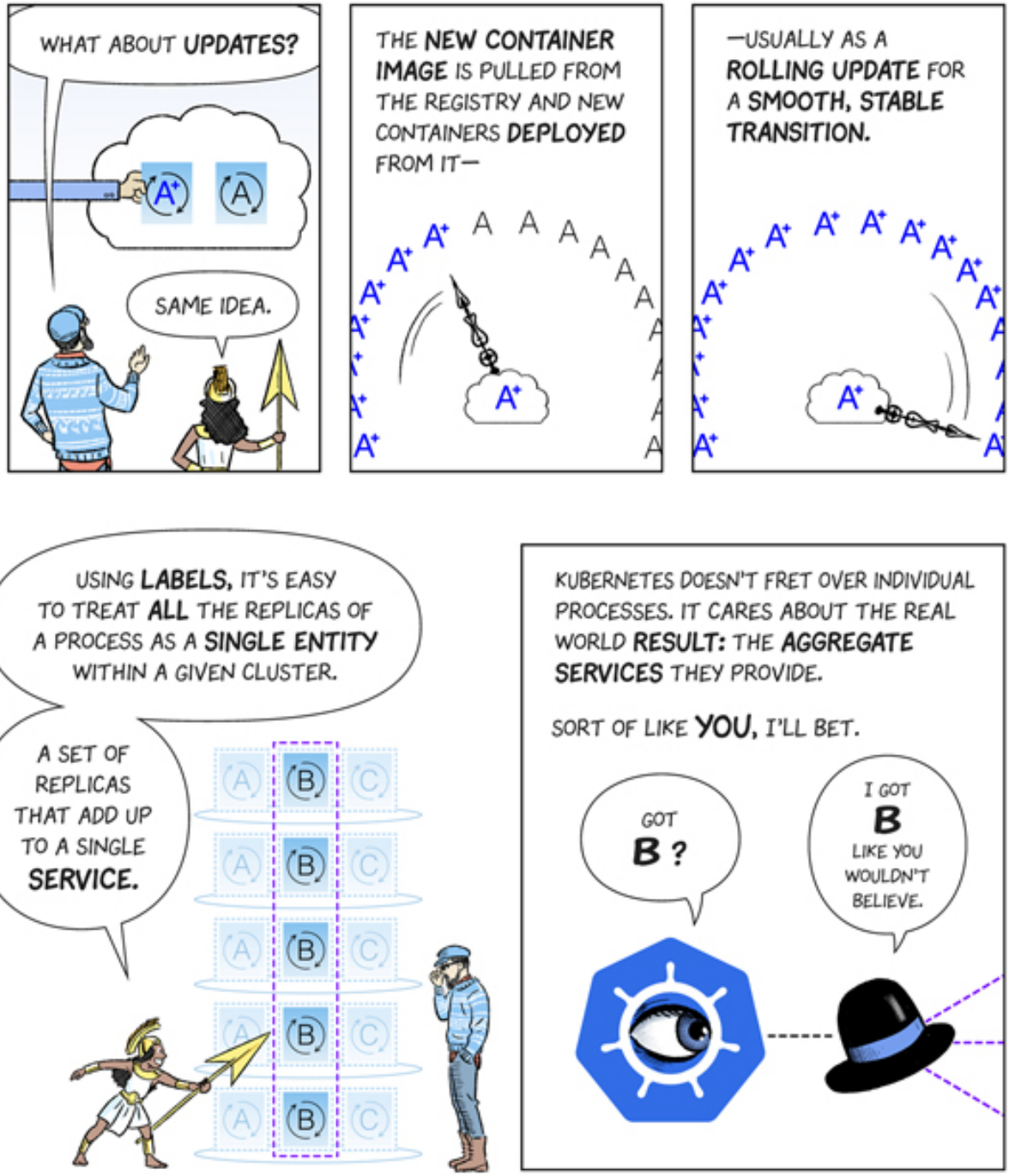

THAT'S THE BEAUTY OF KUBERNETES' ABSTRACTED INFRASTRUCTURE.
WHO CARES WHAT "MACHINE" YOUR PROCESS IS ON? THEY'RE ALL THE SAME TO YOU NOW!

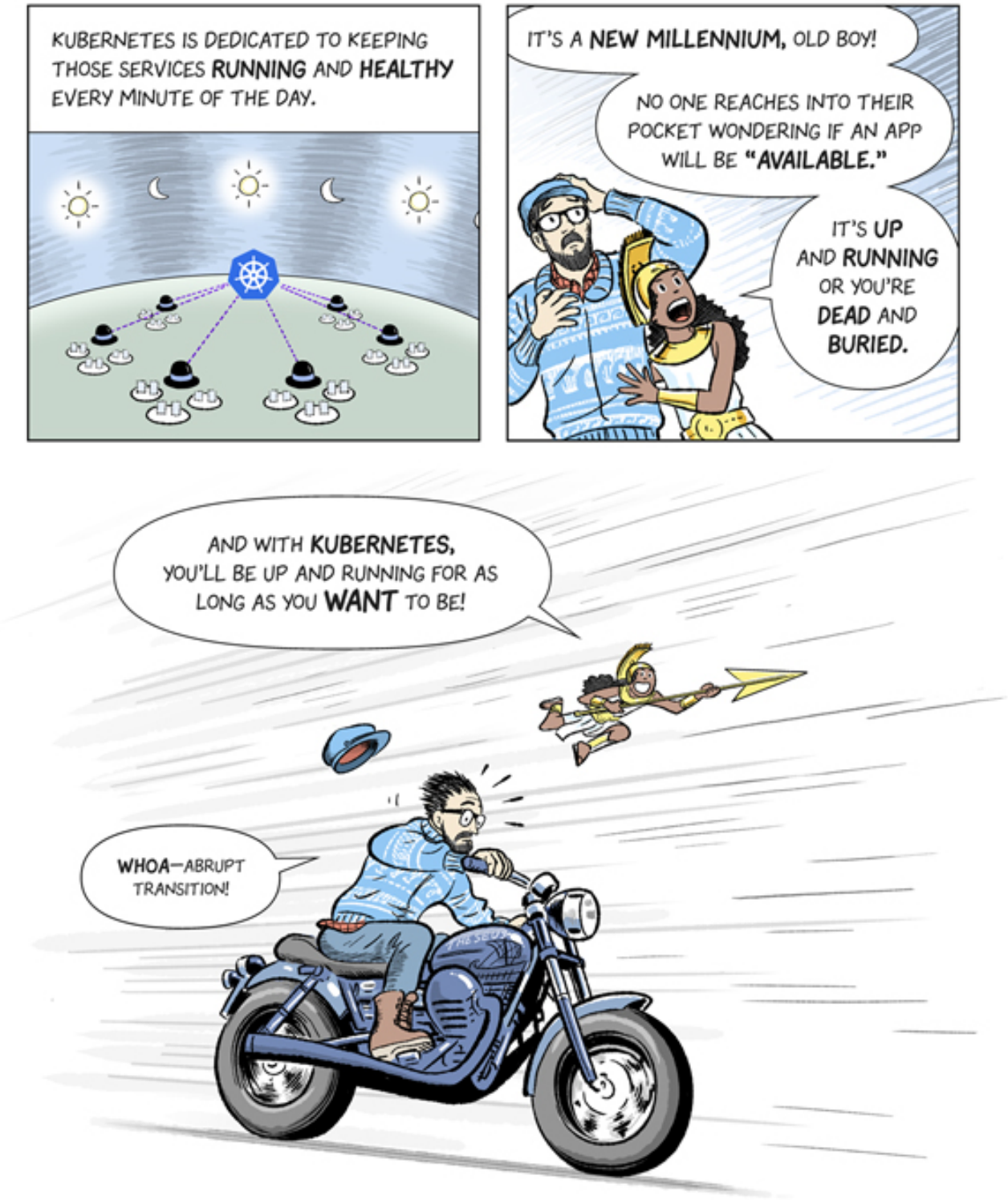

IN THE SAME WAY THAT ALL CONTAINERS MADE FROM A GIVEN IMAGE WILL BE INTERCHANGEABLE WITH ONE ANOTHER.
IDENTICAL, IN FACT, ALL CREATED FROM THE SAME IMMUTABLE TEMPLATE.
A
A
A
A
A

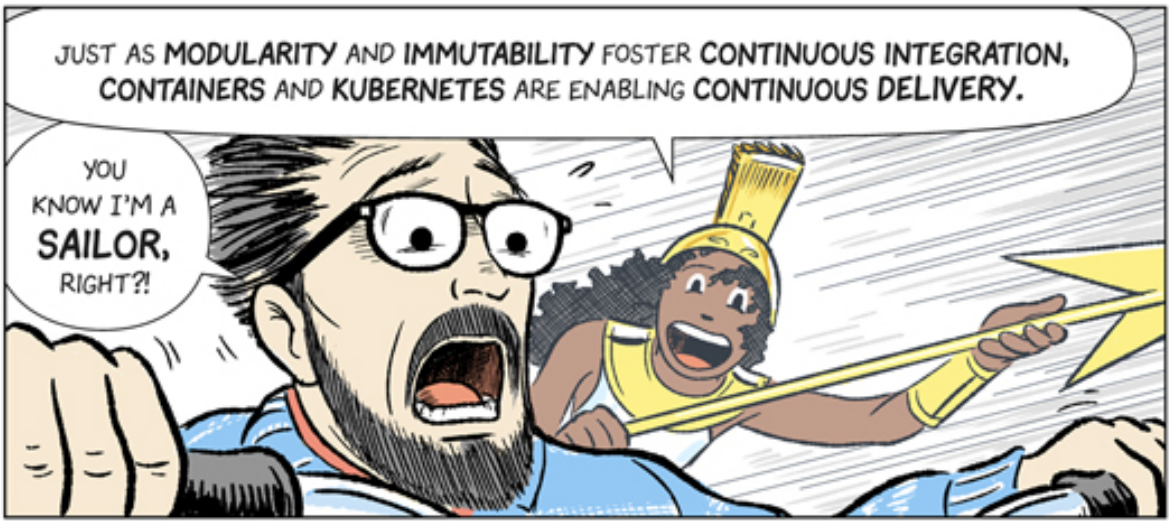

SO, WITH INTERCHANGEABLE CONTAINER REPLICAS AND INTERCHANGEABLE MACHINES—
A
B
C
D

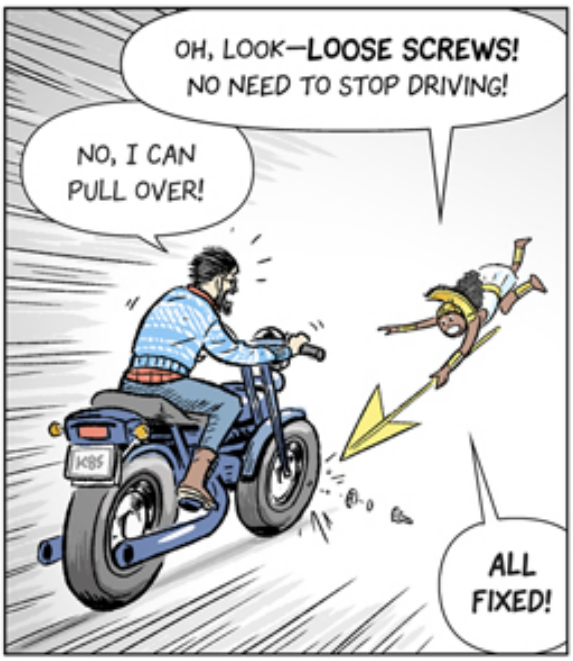

—KUBERNETES CAN DROP EACH REPLICA—
—INTO JUST THE RIGHT PLACE!
TIRED OF—
—WASTING—
—CPU?
KUBERNETES WILL BE ON THE LOOKOUT FOR MORE EFFICIENT BIN PACKING OPPORTUNITIES.
RISK, MEANWHILE, CAN BE DISTRIBUTED, SO THAT NO RUNNING PROCESS—
—CAN BE TAKEN DOWN BY A SINGLE FAILURE POINT.
REPLICAS CAN CONTINUE RUNNING SMOOTHLY, WHILE THE SYSTEM—
—QUICKLY RESTORES THE IDEAL STATE OF YOUR DEPLOYMENT.

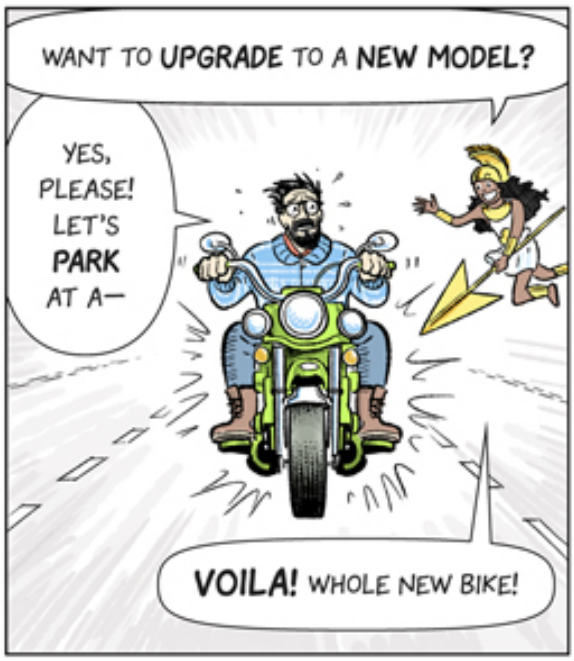

KUBERNETES IS SELF-HEALING.
THE SYSTEM FANATICALLY COMPARES THE IDEAL STATE AS EXPRESSED IN THE DEPLOYMENT—
—TO THE ACTUAL STATE OF PODS AND CLUSTERS IN REAL WORLD OPERATION.
NO TOLERANCE. NO EXCUSES.
IF ANY VIOLATION OR INCONSISTENCY IS DETECTED...
BAM.
B
C
A
B
C
SOMETHING'S GONNA BE TERMINATED—
—AND INSTANTLY REBORN.
A


WHAT ABOUT UPDATES?
SAME IDEA.
THE NEW CONTAINER IMAGE IS PULLED FROM THE REGISTRY AND NEW CONTAINERS DEPLOYED FROM IT—
—USUALLY AS A ROLLING UPDATE FOR A SMOOTH, STABLE TRANSITION.
USING LABELS, IT'S EASY TO TREAT ALL THE REPLICAS OF A PROCESS AS A SINGLE ENTITY WITHIN A GIVEN CLUSTER.
A SET OF REPLICAS THAT ADD UP TO A SINGLE SERVICE.
KUBERNETES DOESN'T FRET OVER INDIVIDUAL PROCESSES. IT CARES ABOUT THE REAL WORLD RESULT: THE AGGREGATE SERVICES THEY PROVIDE.
SORT OF LIKE YOU, I'LL BET.
GOT B ?
I GOT B LIKE YOU WOULDN'T BELIEVE.

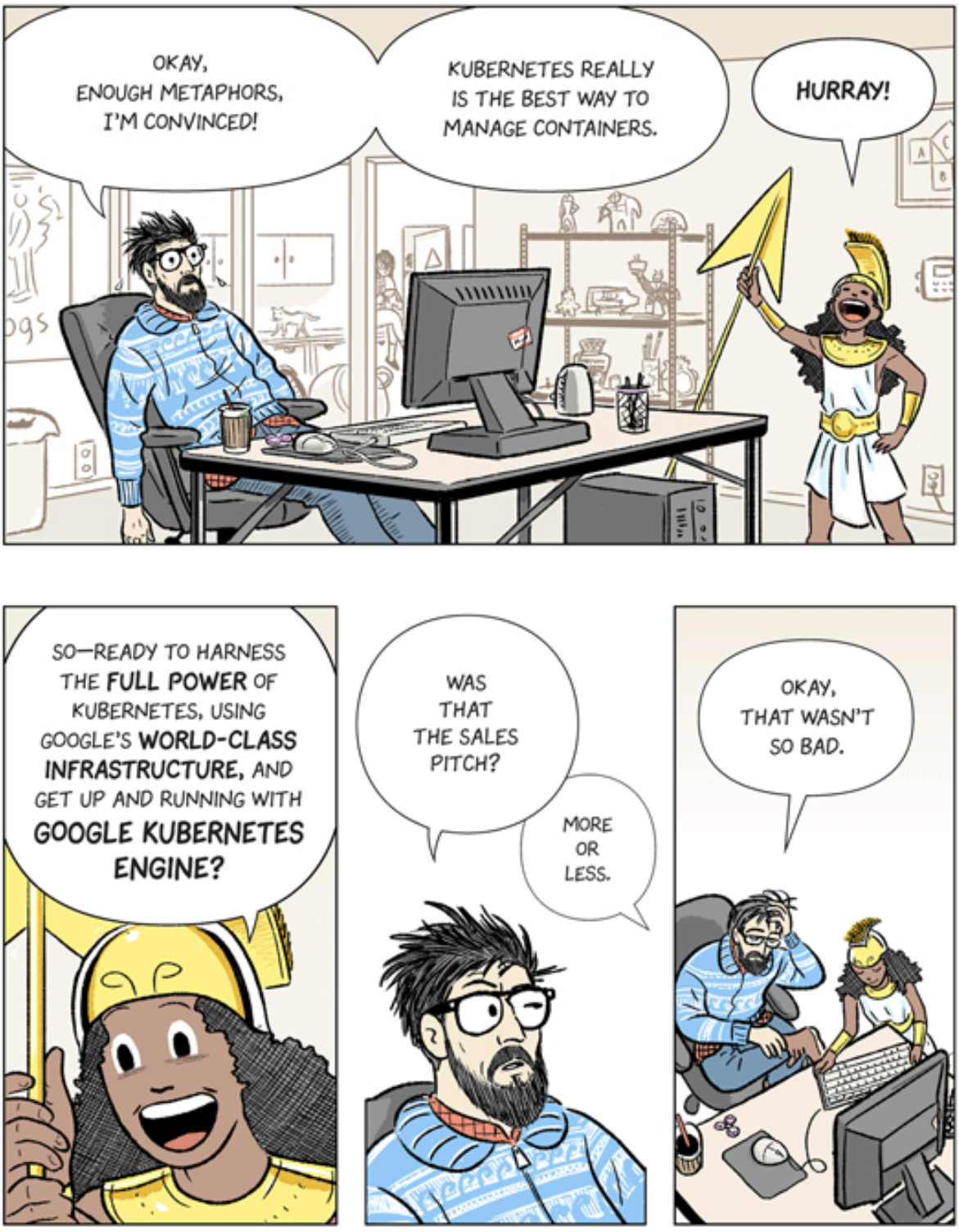

KUBERNETES IS DEDICATED TO KEEPING THOSE SERVICES RUNNING AND HEALTHY EVERY MINUTE OF THE DAY.
IT'S A NEW MILLENNIUM, OLD BOY!
NO ONE REACHES INTO THEIR POCKET WONDERING IF AN APP WILL BE "AVAILABLE."
IT'S UP AND RUNNING OR YOU'RE DEAD AND BURIED.
AND WITH KUBERNETES, YOU'LL BE UP AND RUNNING FOR AS LONG AS YOU WANT TO BE!
WHOA—ABRUPT TRANSITION!







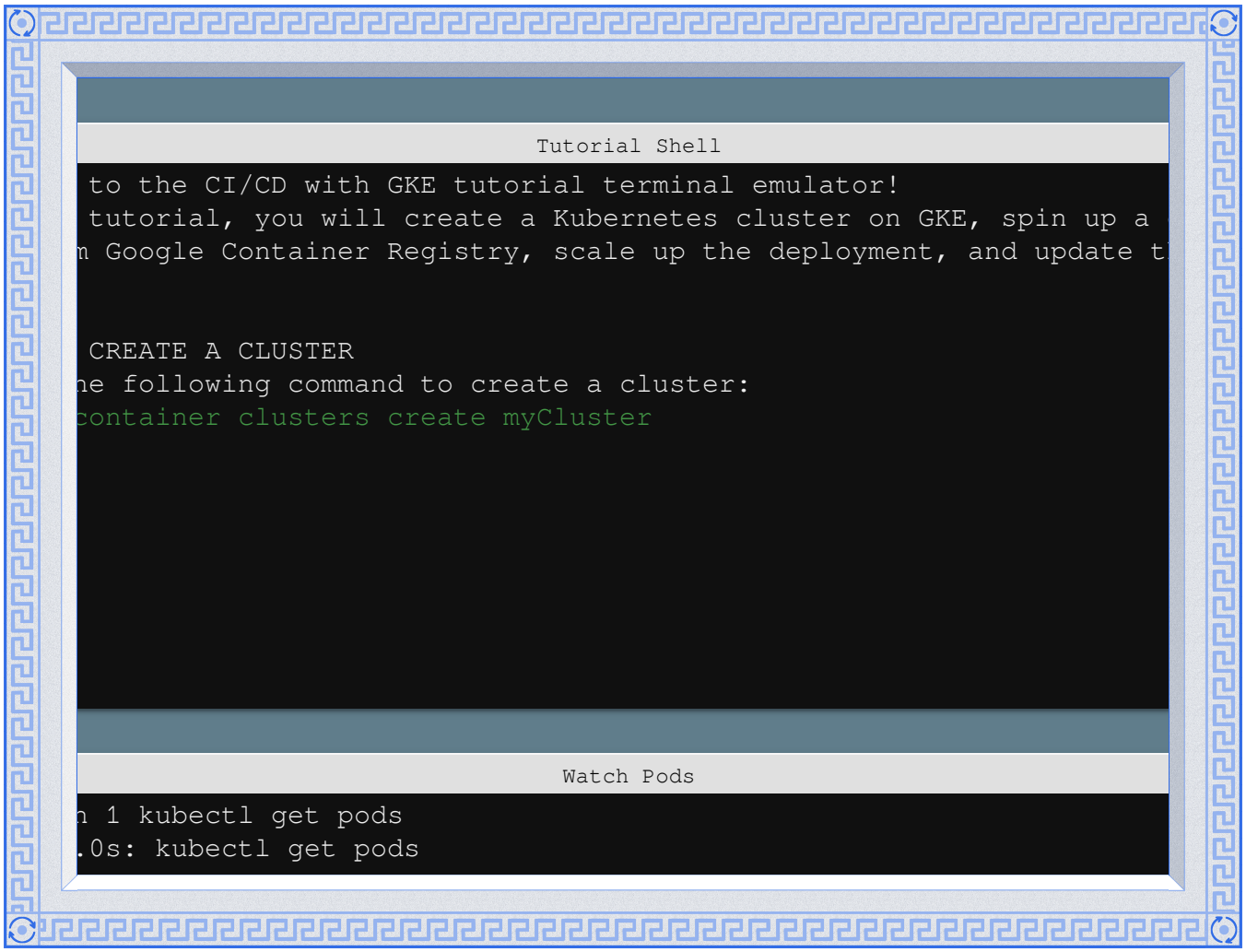
SIX HOURS LATER...


OKAY, ENOUGH METAPHORS, I'M CONVINCED!
KUBERNETES REALLY IS THE BEST WAY TO MANAGE CONTAINERS.
HURRAY!
SO—READY TO HARNESS THE FULL POWER OF KUBERNETES, USING GOOGLE'S WORLD-CLASS INFRASTRUCTURE, AND GET UP AND RUNNING WITH GOOGLE KUBERNETES ENGINE?
WAS THAT THE SALES PITCH?
MORE OR LESS.
OKAY, THAT WASN'T SO BAD.





NOW IT'S YOUR TURN. TYPE COMMANDS INTO THE FOLLOWING TERMINAL EMULATOR AND LEARN HOW TO CREATE A KUBERNETES CLUSTER ON GOOGLE KUBERNETES ENGINE.



ART AND "STORY" BY SCOTT MCCLOUD

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